UNIVERSITY OF PLYMOUTH
School of Psychology

BRIC
BRAIN RESEARCH & IMAGING CENTRE

Participant Information Sheet

Title of Research: Ultrasound neuromodulation of mood, cognition, and reward circuits in mental health

We would like to invite you to take part in our research study.

- Below you will find information about the study, why this research is being conducted and what it would involve. Please take your time to read the information carefully to help you decide whether you would like to take part.
- If you have any questions or need further clarification after reading this information sheet, please do not hesitate to contact the principal investigator via email (details at the end of this document).

What is the study about?

Neuromodulation refers to methods that change how the brain works, and these techniques could potentially help treat mental health issues like depression. However, many of the current methods have not been successful because they do not fully match the brain's complex and varied structure. A new approach called *transcranial ultrasound stimulation (TUS)* is promising because it uses sound waves that can travel through the brain with great precision. This means that TUS can target very specific, small areas (down to a few millimetres) and can make rapid adjustments (within about one-thousandth of a second). These features make TUS a promising candidate for more effective treatments of mental health conditions in the future.

In this study, we aim to investigate how transcranial ultrasound stimulation (TUS) targeted at specific brain regions affects motivation, mood and decision-making in healthy adults.

Who can take part?

You can take part if you are aged 21–65, have normal or corrected-to-normal vision and hearing, and have no current or past diagnosis of a mental health or neurological condition.

What does the study involve?

- **Transcranial Ultrasound Stimulation (TUS):**
  - This safe, non-invasive technique uses low-intensity ultrasound waves to target specific brain regions, temporarily and reversibly modulating brain activity with high precision.
  - You can find out more about how we use TUS in our research lab and view a short video summarising the technique on our webpage: https://www.plymouth.ac.uk/tus
- **Magnetic Resonance Imaging (MRI):**
  - This safe, non-invasive technique requires you to lie down in an MRI scanner and uses a very strong magnetic field to take pictures of your brain
  - **NOTE:** To ensure we can collect high-quality brain images, you must keep your head as still as possible during the scan. To assist with this, foam pads will be placed around your head for support. However, if you believe you will have difficulty staying still, such as due to claustrophobia or general restlessness, please contact the researcher to discuss your suitability for participation.
  - If at any point during the scan you feel distressed or need assistance, you will be able to squeeze a buzzer to alert the radiographer and researcher, who will communicate with you and stop the procedure if needed.
- **Behavioural Task**
  - You will complete a task of decision-making. In the task, you will pursue a goal (filling a net with seafood) by selecting one out of three available resources (*e.g.*, a fish). You will be asked to report how you feel at different instances in the task.

- During the task, we will collect physiological data (heart rate, breathing, sweat response) using disposable electrodes placed on your chest and hand, and a soft band around your chest. We will also record eye movements and pupil size using an eye-tracking system. To help ensure accurate eye-tracking data, you will be asked to rest your chin on a fixed chin rest during the task.
- We will also record your facial movements during the task. **We will not store identifiable facial video images**. Instead, we will store derived facial landmark/mesh data (for example, the tracked position of features such as the eyes, nose, and mouth over time).

- **Mental Health Questionnaires**
  - On a tablet or computer, you will complete questionnaires about your emotions and mood, your health, and your lifestyle. Some questions are optional. At each visit you will also answer a few short questions about how you are feeling that day.
  - Please note that some of these questionnaires contain sensitive topics about **your emotions and mood**, so please consider carefully whether you would feel comfortable reflecting and providing information about these aspects of your mental health.
  - All questionnaires use pre-defined questions and response scales, and your answers will remain anonymous.

What would be expected of me as a participant?

The study will involve **four sessions** spaced at least one week apart at the **Brain Research Imaging Centre (BRIC), Science Park, Plymouth, PL6 8BU**. Please do not take part in the study if you do not think you can commit to attending all sessions.

You should not smoke, vape or drink any alcohol on the day you attend any of the sessions. Caffeinated drinks must be limited to one cup on the morning of your visit. You should not take any other drugs, including additional painkillers, such as Paracetamol, Aspirin and Ibuprofen.

Online Pre-screening Survey (10–15 minutes) – Follow specified link in email

- You will fill in a **TUS and MRI screening questionnaire** as well as a short **mental health questionnaire**. These are used to ensure that it is safe for you to receive brain stimulation and undergo MRI scanning, and to confirm that you meet the study's eligibility criteria related to mental health.
- Please note that participation in this pre-screening is unpaid. We will use this information to determine whether you are eligible for the main study (see below).

Session 1 (75 minutes):

- During your first visit, we will go over the information sheet, explain the procedures, and ask you to re-complete the screening questionnaires to confirm that it is safe for you to take part in the study. We will answer any questions you may have about the study and ask you to sign a consent form.
- **MRI:** A radiographer will go through your screening form with you and answer any queries you may have. We will acquire some images of your brain, requiring you to lay down and rest in the scanner. You will spend around 30 minutes in total in the scanner.
- **Questionnaires:** We will then ask you to complete the questionnaires on a tablet or computer.

Sessions 2-4 (90 minutes each):

- The next sessions will each follow the same procedure. Upon arrival, you will recomplete the screening questionnaires to ensure you are still safe to participate.
- Before each stimulation session, please come with clean, dry hair with no products in it, and avoid wearing eye make-up, as it affects the eye tracking.

- **TUS**: In the Brain Stimulation lab, the researcher will apply a water-based gel to a portion of your head, position the TUS device over this region and deliver pulses of ultrasound for up to 12 minutes in total, targeting different areas of the brain.
- **Behavioural task:** You will complete the behavioural task whilst we record the physiological measures.
- **End of the session:** You will be provided with a towel and shampoo to wash off the gel. We will ask you to complete a questionnaire to report any symptoms you experienced following the stimulation after each study visit.

At the end of the final session, you will be debriefed and compensated £110 for your time and participation, plus up to £10 for each TUS session based on your task performance. Thus, the maximum you could receive is £140.

What will happen if I decide I want to take part?

Please follow the link in the email to complete MRI and TUS screening forms, as well a brief mental health questionnaire. If there are no eligibility or safety concerns regarding your participation, we will schedule your first study visit.

Do I have to take part?

No, it is up to you whether or not to take part. If you decide after reading this information sheet that you do not wish to take part, you can let the researcher know without giving a reason and without any penalty.

What if I decide I no longer wish to partake in the study?

Your participation is voluntary, and you are free to withdraw at any point during the study without giving any reason and without penalty. If you wish to withdraw during the study, please make this known to the researcher. You can withdraw your data up to two weeks after you have completed the study by notifying the principal investigator.

Will it be confidential and anonymous?

The personal information you provide during the study will be stored and processed in accordance with the General Data Protection Regulation and the Data Protection Act 2018. Your experimental data will be fully anonymised using a unique code and it will not be possible to link the data back to your identity. In a separate, password protected secure database we will store your name, email address and unique code, in case you wish to have your data withdrawn. All information collected about you during this study will be kept strictly confidential.

How will my data be used?

During this study, we will collect demographic data (including ethnicity), optional health information, mental health questionnaire scores, brain imaging data, physiological data (heart rate, breathing, sweat response, eye tracking, non-identifiable facial movements) and task performance. Your anonymised data will be aggregated with that from other participants and analysed by the research team. Over the coming years, results may be presented at scientific meetings, published in scientific journals, or published on our website. We may share your data as “open data”. This means your anonymised data will be publicly available to other researchers, inside and outside the European Union and used in future studies for purposes not related to this study. It will not be possible to identify you from any raw or published data. With your consent, we will keep your personal information on a secure database to contact you for future studies.

What are the benefits of taking part in this study?

While there is no direct benefit to you, your participation will contribute to a better understanding of how brain circuits support decision-making. This knowledge could guide future research on the use of TUS as a potential intervention for mental health disorders.

Are there any risks or side effects?

All procedures are non-invasive and safe; you should not experience long-lasting discomfort or side effects. However, please review the following potential risks to help you make an informed decision about participating:

- ***Mental Health Questionnaires:*** Due to the sensitive nature of the questionnaires assessing various aspects of your mental health including your **emotions, mood and health**, it is possible you may experience some emotional distress from answering these questions. If you think this is likely, we ask that you do not continue with the study.
- **TUS:** TUS is **safe and completely non-invasive**, the stimulation parameters we will use adhere with safety guidelines set by the International Transcranial Ultrasonic Stimulation Safety and Standards of which our lab head, Professor Elsa Fouragnan, is an executive board member.
    - TUS has been used on over 200 participants in at BRIC with no side effects reported aside those commonly associated with neuromodulation (e.g. mild headache, which usually responds well to over-the-counter painkillers i.e. paracetamol). As a precaution, if you suffer with **migraines**, you should not take part.
    - TUS will not cause seizures. There has been no report of seizures in more than 100 human studies performed worldwide (including 5 completed TUS studies at BRIC). However, as a precaution, we will not give TUS to someone with a personal or close family history (first-degree relative e.g. parent, sibling, child) of **epilepsy**, or with a personal history of another significant **neurological or psychiatric disorder** or **extreme mood fluctuations**.
    - If you are taking any **medication**, you should discuss this with the researcher beforehand.
    - It is our policy not to give TUS to someone who is **pregnant**. If you are pregnant, think you may be pregnant, or are planning to become pregnant, please do not participate in this study.
- **MRI:** MRI is **safe and non-invasive** and does not involve any ionizing radiation (x-rays). However, because it uses a large magnet to work, MRI scans are not suitable for everybody.
    - **Pre-screening:** You must complete a safety questionnaire to ensure you do not have any metal in your body. If you have a pacemaker, mechanical heart valve, mechanical implant such as an aneurysm clip, hip replacement, or other pieces of metal that has accidentally entered your body, please make this known.
    - **Claustrophobia**: The scan requires you to lie still with your head in an enclosed space. If you believe you may experience claustrophobia, we recommend that you refrain from participating to avoid unnecessary stress.
    - **Noise:** The scans are very noisy, but we will give you earplugs to make this quieter for you.
    - **Clothing/Jewellery/Piercings:** For your comfort during the scan and safety, we ask that you wear comfy clothing with no excessive metal that cannot be removed. Avoid wearing fabrics containing metallic threads or are silver impregnated (often marketed as anti-microbial/bacterial or anti-odour). If you wear a bra, please choose a non-wired one if possible. If there is concern about your clothing, you may be asked to change into hospital scrubs for the scan. Metal jewellery including body piercings must be removed. Eye makeup must also be avoided. Lockers are provided to secure your personal belongings.
    - **These scans are not for diagnostic purposes**, not reviewed by a doctor, and therefore not a substitute for a medical appointment. However, if a possible abnormality is detected, the scan will be sent to be checked by a doctor. If the doctor feels that the abnormality is medically important, you will be contacted directly and recommended to arrange a hospital (NHS) diagnostic scan. You would not be informed unless the doctor thinks the finding has clear implications for your health. All information about you is kept strictly confidential.

Can I withdraw from the study?

Your participation is voluntary, and you are free to withdraw at any point during the study without giving any reason and without penalty, by making this known to the researcher. You can withdraw your data up to two weeks following completion of the study by emailing the principal investigator (see contact details below). You will still be reimbursed for the time that you have spent participating in the study up to the point of withdrawal. Not participating/ withdrawing from the study will have no impact upon your relationship with the University, or associated staff.

Who has reviewed the study?

This project has been reviewed on behalf of the University of Plymouth, Faculty Research Ethics and Integrity Committee, and was given approval with the following Application ID: 6295

Where can I find out more about how my information is used?

Find out more about how we use your information at https://www.plymouth.ac.uk/research/governance/research-participant-privacy-notice, by contacting a member of the research team, or by contacting the University of Plymouth's Data Protection Officer at: dpo@plymouth.ac.uk.

Contact details of researchers:

- ***Principal Investigator:***
  Dr Dylan Curtin
  dylan.curtin@plymouth.ac.uk
- ***Head of Laboratory:***
  Professor Elsa Fouragnan
  elsa.fouragnan@plymouth.ac.uk

If you have any concerns or complaints about the ethical conduct of this study, please contact the Research Administrator, Faculty of Health Research Ethics and Integrity Committee, University of Plymouth, Level 2 Marine Building, Drake Circus, Plymouth PL4 8AA Email: Research.Ethics@plymouth.ac.uk

Thank you for reading this information sheet.